

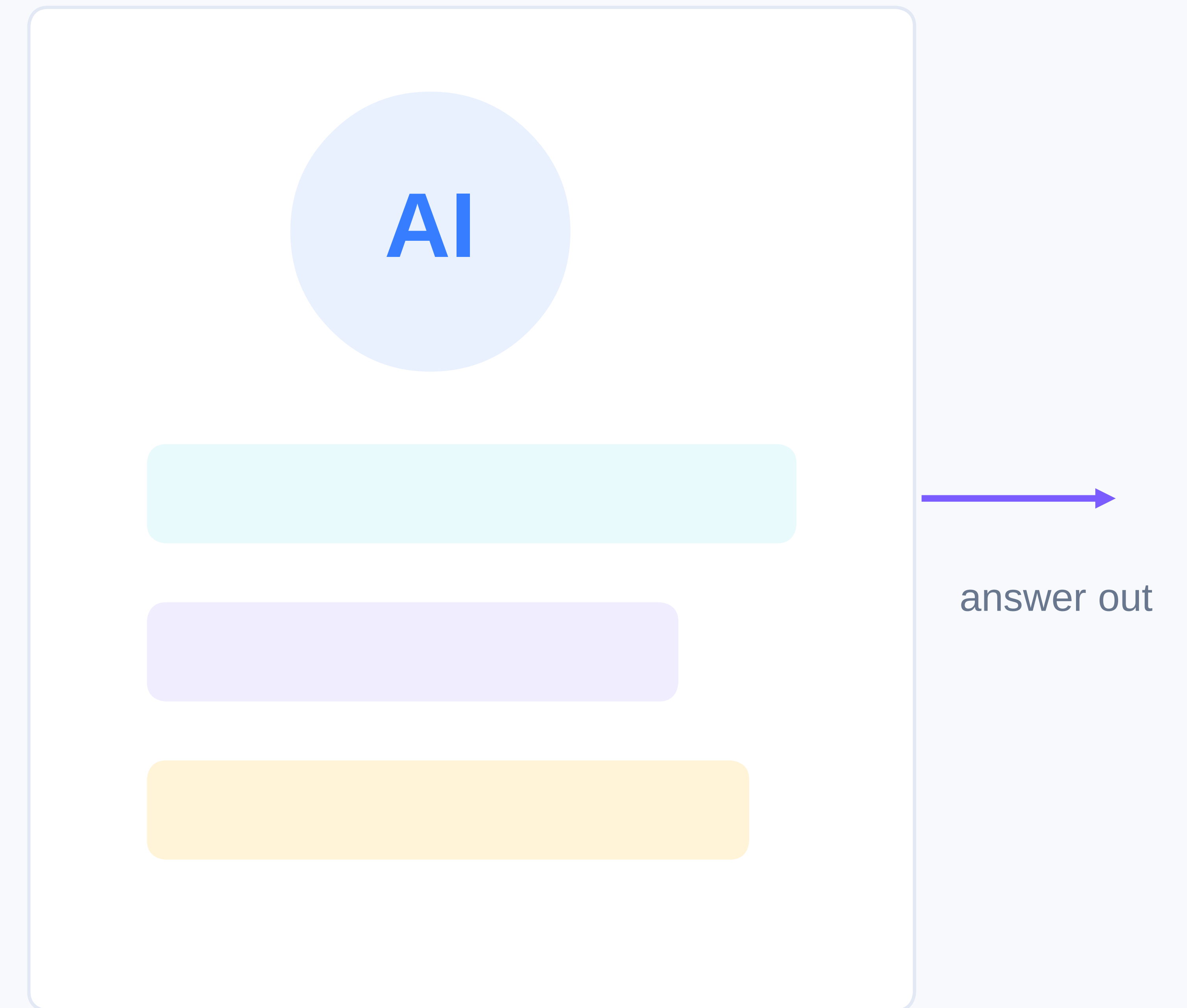
AI Transformer Models

A simple, visual explanation for non-technical audiences

No math required

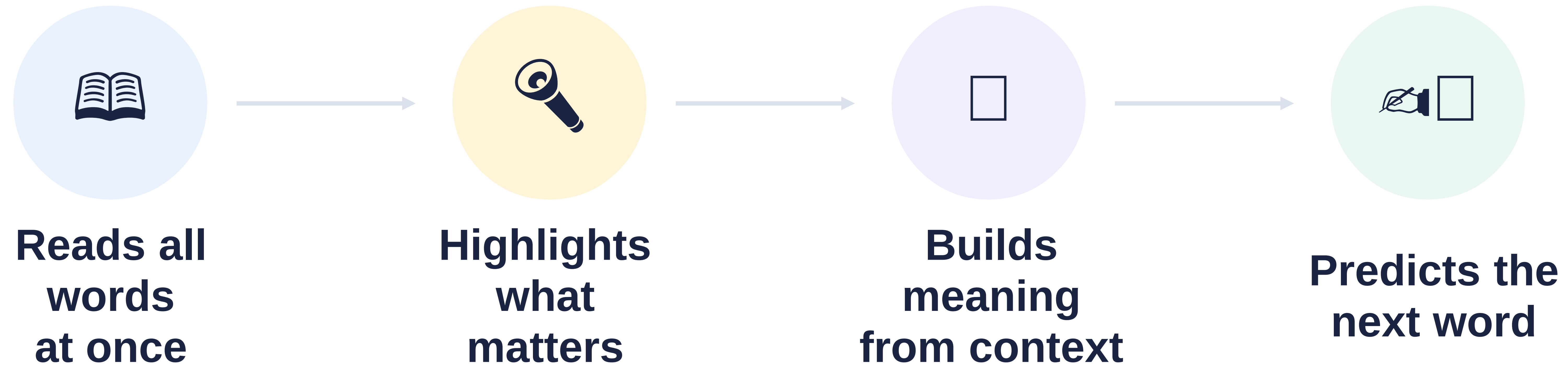
Think: smart reading

words in



The big idea

A transformer is like a very fast reader with many highlighters.



Unlike older models that read left-to-right slowly, transformers compare many words at the same time.

Step 1: Split text into small pieces

The model first turns a sentence into tokens — small chunks of text.

Example sentence

The robot picked up the red block



The

#1

robot

#2

picked

#3

up

#4

the

#5

red

#6

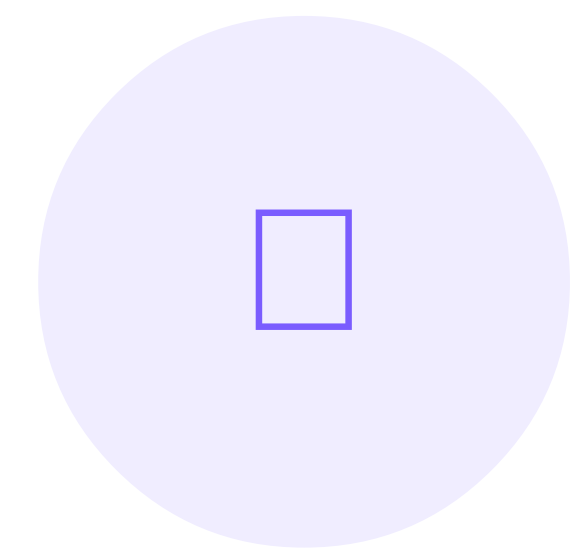
block

#7

Why? Computers work with numbers, not words. Tokens are the bridge from text to numbers.

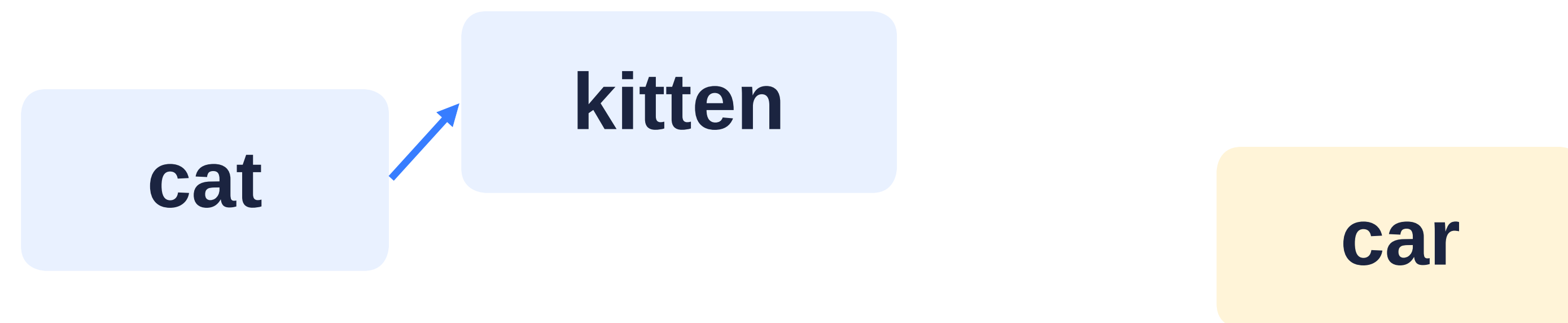
Step 2: Give each token meaning + position

A transformer needs to know both what a word means and where it appears.



Meaning

Words with similar meanings get similar number patterns.



Position

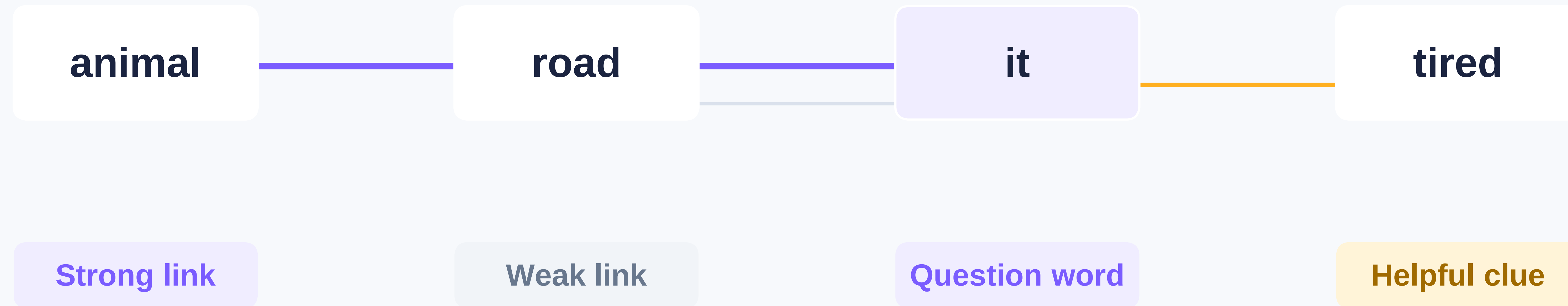
Order changes meaning. “Dog bites man” is not the same as “Man bites dog.”



Step 3: Attention asks “what should I look at?”

Each token checks which other tokens are most useful for understanding it.

The animal did not cross the road because it was tired.



Attention helps the model connect “it” to the right idea in the sentence.

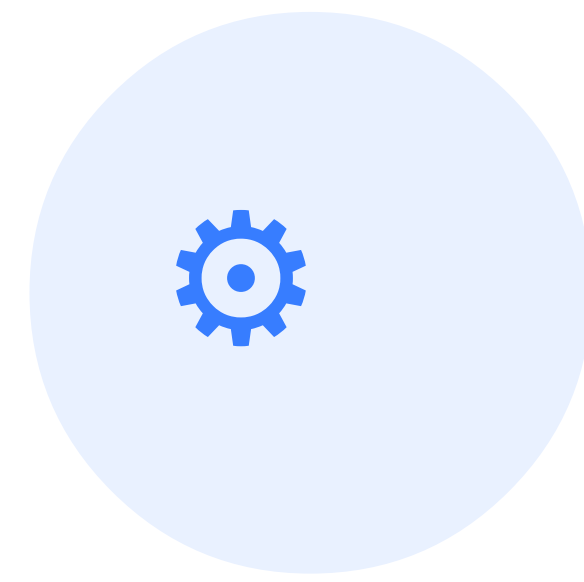
Step 4: Attention heads find different clues

One head may track grammar; another may track meaning; another may track long-distance links.



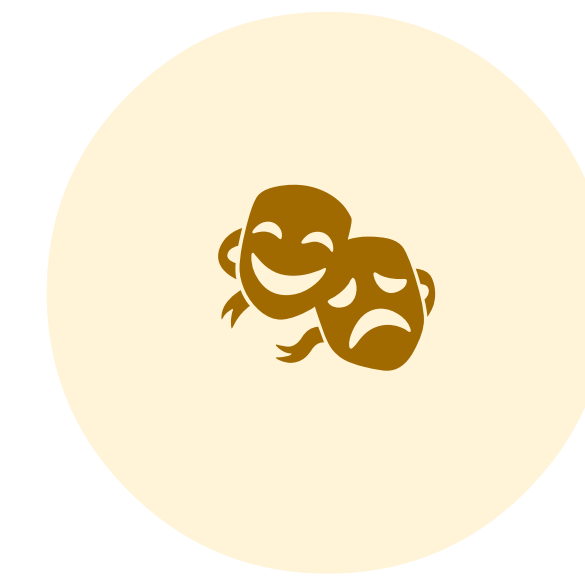
Head A

Who does “it” refer to?



Head B

Which words describe
action?



Head C

What is the emotional
tone?



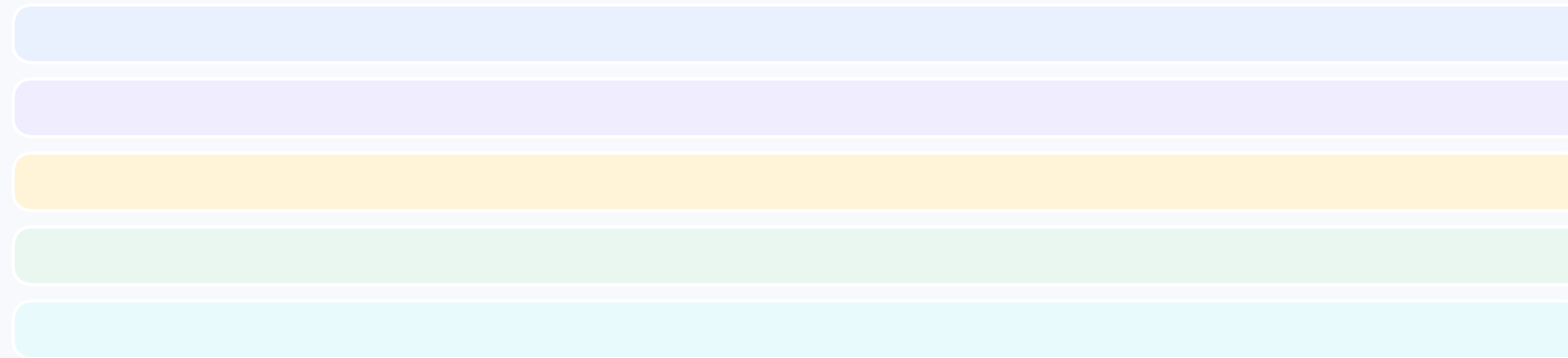
The results are combined into a richer understanding of the sentence.

A transformer layer: compare, mix, clean up

The model repeats this layer many times to build deeper context.



Layer stack

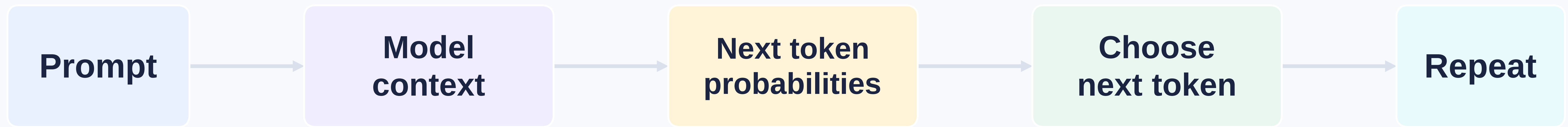


Layer 1 → Layer 2 → Layer 3 → Layer 4 → Layer 5

Repeating layers lets the model handle complex instructions and long context.

How it produces an answer

Most text transformers generate one token, then use it to choose the next token.



“The best way to improve quality is to ...”

measure

inspect

guess

The model does not “look up” the answer; it estimates likely useful continuations from patterns learned during training.

Strengths and cautions

They are powerful pattern learners, not perfect truth machines.

Good at

- Summarising text
- Answering questions
- Drafting and rewriting
- Finding patterns in data

Be careful with

- Making up confident answers
- Outdated knowledge
- Sensitive or private data
- Tasks needing expert review

Best use: combine AI speed with human judgment.

Recap: the transformer in 5 simple steps

From text input to useful output.



Simple mental model

“A transformer is a smart text processor that repeatedly checks context and predicts what should come next.”